MODEL LAB · 2026-10-17

# Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

**Served (full season)**
served prior

**11**
priors compared

**26,271**
held-out plate appearances

**3.07%**
held-out HR rate

**0.6059**
best AUC · KNN comps

## Held-out scores

Fitted on plate appearances before 2026-09-04, scored on the 26,271 that came after (actual HR rate 3.07%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

| PRIOR | AUC | TOP-DECILE LIFT | BRIER | LOG LOSS | CAL. RMSE |
|---|---|---|---|---|---|
| **Flat league**<br>League HR rate as the prior for every hitter. | 0.5794 | 1.65× | 0.02968 | 0.13619 | 0.00571 |
| **KNN comps**<br>Prior is the pooled rate of the nearest comparables. | 0.6059 | 1.56× | 0.02962 | 0.13488 | 0.00390 |
| **SVR**<br>Support-vector regression on the same feature matrix. | 0.5972 | 1.56× | 0.02967 | 0.13558 | 0.00596 |
| **Random forest**<br>Forest regression; a learned similarity metric. | 0.5982 | 1.55× | 0.02964 | 0.13516 | 0.00523 |
| **Core (KNN+SVR)**<br>Blend of KNN and SVR, no forest. | 0.6059 | 1.56× | 0.02962 | 0.13488 | 0.00390 |
| **Forest blend**<br>KNN + SVR + forest, weights fitted out-of-fold. | 0.6059 | 1.56× | 0.02962 | 0.13488 | 0.00390 |
| **Linear regression**<br>Ordinary least squares on the same features. | 0.6017 | 1.58× | 0.02965 | 0.13518 | 0.00398 |
| **Logistic regression**<br>Binomial logistic on the per-PA home run outcome. | 0.6027 | 1.65× | 0.02966 | 0.13524 | 0.00623 |
| **Neural net (torch)**<br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.6007 | 1.60× | 0.02964 | 0.13512 | 0.00470 |
| **XGBoost**<br>Gradient-boosted trees, depth 3 with subsampling and L2. | 0.5988 | 1.58× | 0.02966 | 0.13527 | 0.00582 |

### BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

### SHRINKAGE K PER PRIOR

league **251** knn **251** svr **251** rf **251** core **251** ensemble **251** linear **251** logistic **251** nn **251** xgb **251**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

## Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (2,786 PA, actual HR rate 2.37%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

| PRIOR | AUC | TOP-DECILE LIFT | BRIER | LOG LOSS | CAL. RMSE |
|---|---|---|---|---|---|
| **Flat league**<br>League HR rate as the prior for every hitter. | 0.5860 | 1.36× | 0.02314 | 0.11229 | 0.01037 |
| **KNN comps**<br>Prior is the pooled rate of the nearest comparables. | 0.6232 | 1.97× | 0.02306 | 0.10992 | 0.00977 |
| **SVR**<br>Support-vector regression on the same feature matrix. | 0.6017 | 1.82× | 0.02325 | 0.11321 | 0.01716 |
| **Random forest**<br>Forest regression; a learned similarity metric. | 0.6106 | 2.12× | 0.02316 | 0.11068 | 0.01542 |
| **Core (KNN+SVR)**<br>Blend of KNN and SVR, no forest. | 0.6232 | 1.97× | 0.02306 | 0.10992 | 0.00977 |
| **Forest blend**<br>KNN + SVR + forest, weights fitted out-of-fold. | 0.6232 | 1.97× | 0.02306 | 0.10992 | 0.00977 |
| **Linear regression**<br>Ordinary least squares on the same features. | 0.6211 | 1.97× | 0.02326 | 0.11135 | 0.01418 |
| **Logistic regression**<br>Binomial logistic on the per-PA home run outcome. | 0.6307 | 1.66× | 0.02325 | 0.11095 | 0.01676 |
| **Neural net (torch)**<br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.6236 | 1.82× | 0.02308 | 0.11000 | 0.01268 |
| **XGBoost**<br>Gradient-boosted trees, depth 3 with subsampling and L2. | 0.6196 | 1.82× | 0.02326 | 0.11125 | 0.01972 |

## Slate agreement

| PRIOR | SHARED | ITS TOP SIX |
|---|---|---|
| **Served (full season)** | 6/6 | Jonathan Aranda 12.8%, Jo Adell 12.3%, Junior Caminero 12.1%, Patrick Bailey 11.6%, Chase DeLauter 10.6%, Victor Mesa Jr. 10.2% |
| **Flat league** | 5/6 | Jonathan Aranda 12.1%, Jo Adell 11.6%, Junior Caminero 11.5%, Patrick Bailey 11.0%, Chase DeLauter 10.7%, Travis Bazzana 10.1% |
| **KNN comps** | 6/6 | Jonathan Aranda 12.9%, Jo Adell 12.0%, Junior Caminero 12.0%, Patrick Bailey 11.7%, Chase DeLauter 10.7%, Victor Mesa Jr. 10.3% |
| **SVR** | 6/6 | Junior Caminero 13.1%, Jo Adell 12.5%, Jonathan Aranda 12.4%, Patrick Bailey 11.5%, Victor Mesa Jr. 10.8%, Chase DeLauter 10.4% |
| **Random forest** | 6/6 | Jonathan Aranda 12.7%, Junior Caminero 12.3%, Jo Adell 12.2%, Patrick Bailey 11.2%, Victor Mesa Jr. 10.7%, Chase DeLauter 10.6% |
| **Core (KNN+SVR)** | 6/6 | Jonathan Aranda 12.9%, Jo Adell 12.0%, Junior Caminero 12.0%, Patrick Bailey 11.7%, Chase DeLauter 10.7%, Victor Mesa Jr. 10.3% |
| **Forest blend** | 6/6 | Jonathan Aranda 12.9%, Jo Adell 12.0%, Junior Caminero 12.0%, Patrick Bailey 11.7%, Chase DeLauter 10.7%, Victor Mesa Jr. 10.3% |
| **Linear regression** | 6/6 | Jonathan Aranda 12.7%, Jo Adell 12.2%, Junior Caminero 12.1%, Patrick Bailey 11.4%, Victor Mesa Jr. 10.5%, Chase DeLauter 10.4% |
| **Logistic regression** | 6/6 | Jo Adell 12.6%, Jonathan Aranda 12.5%, Junior Caminero 12.5%, Patrick Bailey 11.1%, Victor Mesa Jr. 10.5%, Chase DeLauter 10.3% |
| **Neural net (torch)** | 6/6 | Jonathan Aranda 12.6%, Junior Caminero 12.5%, Jo Adell 12.5%, Patrick Bailey 11.3%, Victor Mesa Jr. 10.6%, Chase DeLauter 10.3% |
| **XGBoost** | 6/6 | Junior Caminero 12.9%, Jonathan Aranda 12.7%, Jo Adell 12.4%, Patrick Bailey 11.1%, Victor Mesa Jr. 10.4%, Chase DeLauter 10.4% |

## Hitter by hitter

The top 40 bats by the served probability.

| | HITTER | | SERVED (FULL SEASON) | FLAT LEAGUE | KNN COMPS | SVR | RANDOM FOREST | CORE (KNN+SVR) | FOREST BLEND | LINEAR REGRESSION | LOGISTIC REGRESSION | NEURAL NET (TO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **Jonathan Aranda** | TB slate | **12.8%** | 12.1% | 12.9% | 12.4% | 12.7% | 12.9% | 12.9% | 12.7% | 12.5% | |
| 2 | **Jo Adell** | CLE slate | **12.3%** | 11.6% | 12.0% | 12.5% | 12.2% | 12.0% | 12.0% | 12.2% | 12.6% | |
| 3 | **Junior Caminero** | TB | **12.1%** | 11.5% | 12.0% | 13.1% | 12.3% | 12.0% | 12.0% | 12.1% | 12.5% | |
| 4 | **Patrick Bailey** | CLE | **11.6%** | 11.0% | 11.7% | 11.5% | 11.2% | 11.7% | 11.7% | 11.4% | 11.1% | |
| 5 | **Chase DeLauter** | CLE | **10.6%** | 10.7% | 10.7% | 10.4% | 10.6% | 10.7% | 10.7% | 10.4% | 10.3% | |
| 6 | **Victor Mesa Jr.** | TB | **10.2%** | 10.0% | 10.3% | 10.8% | 10.7% | 10.3% | 10.3% | 10.5% | 10.5% | |
| 7 | **Travis Bazzana** | CLE | **10.2%** | 10.1% | 10.1% | 9.8% | 9.9% | 10.1% | 10.1% | 10.0% | 9.7% | |
| 8 | **Nathaniel Lowe** | CLE | **10.0%** | 9.9% | 10.2% | 10.0% | 10.1% | 10.2% | 10.2% | 10.0% | 10.0% | |
| 9 | **Yandy Díaz** | TB | **9.6%** | 9.9% | 9.7% | 10.1% | 9.8% | 9.7% | 9.7% | 9.8% | 9.6% | |
| 10 | **José Ramírez** | CLE | **9.1%** | 9.3% | 9.2% | 9.3% | 9.3% | 9.2% | 9.2% | 9.5% | 9.7% | |
| 11 | **Petey Halpin** | CLE | **8.3%** | 9.2% | 8.3% | 8.7% | 8.5% | 8.3% | 8.3% | 9.3% | 8.9% | |
| 12 | **Brayan Rocchio** | CLE | **6.3%** | 6.8% | 6.2% | 6.3% | 5.9% | 6.2% | 6.2% | 6.0% | 6.0% | |
| 13 | **Liam Hicks** | TB | **6.1%** | 7.0% | 6.1% | 6.9% | 6.0% | 6.1% | 6.1% | 6.5% | 6.3% | |
| 14 | **Jonny DeLuca** | TB | **5.6%** | 6.1% | 5.5% | 5.5% | 5.2% | 5.5% | 5.5% | 5.5% | 5.7% | |
| 15 | **Richie Palacios** | TB | **5.4%** | 6.2% | 5.5% | 5.3% | 5.0% | 5.5% | 5.5% | 5.4% | 5.4% | |
| 16 | **Steven Kwan** | CLE | **4.4%** | 7.4% | 4.7% | 3.1% | 3.6% | 4.7% | 4.7% | 3.0% | 4.4% | |
| 17 | **Taylor Walls** | TB | **3.6%** | 5.1% | 3.7% | 3.0% | 3.2% | 3.7% | 3.7% | 3.8% | 3.8% | |
| 18 | **Chandler Simpson** | TB | **2.5%** | 5.4% | 3.1% | 1.8% | 1.4% | 3.1% | 3.1% | 1.4% | 2.8% | |

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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Slate for 2026-10-17 · model 2026.10.01-stack · built 2026-10-18 04:22:05 · live data 2026-10-18 04:22:05

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